Centers for Medicare & Medicaid Services
Assistant Secretary for Technology Policy (ASTP)/
Office of the National Coordinator for Health Information Technology (ONC)
Department of Health and Human Services
Attention: CMS-0042-NC
Mail Stop C4-26-05
7500 Security Boulevard
Baltimore, MD 21244-1850

RE: Health Technology Ecosystem Request for Information (RFI)

Thank you for the opportunity to comment on the referenced Request for Information (RFI). The HL7® Da Vinci Project and its membership strongly support driving large-scale adoption of health technology applications, reducing barriers to data access and exchange, and leveraging technology and innovation to support a move to value-based care. As an HL7 Accelerator, the Da Vinci Project is responsible for supporting industry, specifically providers and payers, but also critical vendor partners, to develop and implement value-based care standards. We work together to support better healthcare at lower costs. In this response we focus on how supporting and accelerating the use of HL7 Fast Healthcare Interoperability Resources (FHIR®)-based Application Programming Interfaces (APIs) and continuing to advance interoperability, actionable data exchange, and transparency can help CMS and ASTP/ONC meet their goals, achieve HHS priorities, and ultimately improve patient healthcare outcomes.

CMS laid a solid foundation for FHIR-based innovation and efficiency with the CMS Interoperability and Patient Access Final Rule (CMS-9115-F) and the CMS Interoperability and Prior Authorization Final Rule (CMS-0057-F). This was supported by the ONC 21st Century Cures Act Final Rule and further advanced through certification efforts supported by the ASTP/ONC Health Data, Technology, and Interoperability: Certification Program Updates, Algorithm Transparency, and Information Sharing (HTI-1) Final Rule. It is on this foundation that the Da Vinci Project believes CMS and ASTP/ONC can continue to build Federal interoperability and inter-agency collaboration, and signal to industry the importance of continuing to move this work forward. Industry can use these regulations to innovate and progress through community-developed, consensus-based standards to improve healthcare.

While all topics covered in this RFI are important, the Da Vinci Project prioritizes areas where we, as an HL7 Accelerator consisting of payers, providers, and vendors, can most effectively contribute to advancing this work. In the following sections, we address issues related to provider data exchange; payers and their role in promoting interoperability; the opportunities for technology vendors, data providers, and networks to support an innovative and efficient health IT ecosystem for patients; and the importance of standards-based APIs in achieving value-based care and system efficiencies at scale.

II.C. Providers –

2. Data Exchange

PR-5 – Use of FHIR APIs, Capabilities, & Constraints: Da Vinci Project member organizations are actively working to implement many of these APIs and standards. As with the rest of the industry, they are at different levels of engagement from development, to testing, to production.

Patient Access API: CMS-9115-F called out the HL7 FHIR Da Vinci Payer Data Exchange (PDex) Implementation Guide (IG) and PDex Drug Formulary IG as IGs to help facilitate implementing this API. We continue to work to advance the PDex IG to ensure it cannot only support third-party apps making data available to patients generally, but can also support new requirements in CMS-0057-F to make Prior Authorization (PA) data available via the Patient Access API. Patients can greatly benefit from having access to their data in a format they can best use and understand through an app of their choice. But implementation and adoption of the Patient Access API has been limited. One challenge is that many patients do not know apps are available to them and app developers and payers need motivation to push adoption. There is an opportunity for CMS and ONC/ASTP to support publicizing Patient Access API-powered apps, and payer engagement with them, to ensure patients are getting the greatest benefit from this API.

Provider Access API and **Payer to Payer API**: CMS also recommended PDex be used to support the Provider Access API to support payer to provider data sharing and for the Payer to Payer API to support data sharing between payers. There is significant value in being able to use the same suite of IGs for both APIs as well as the Patient Access API as leveraging the same standards across use cases can limit implementation burden. And, there is great value in a patient's data moving with them as they engage with new providers and move between payers. There are challenges with attribution and consent, however.

The Da Vinci Member Attribution (ATR) List IG, as detailed in the PDex IG, is foundational to value-based care and supports attribution and population identity. CMS and ASTP/ONC can provide additional guidance to support how to help patients move the data they wish to move to only those providers they wish to see it. Granular opt-out/opt-in could enhance patient engagement. Additionally, a standardized approach to consent that is widely applied across the health IT ecosystem could also support patient engagement and data sharing. CMS and ASTP/ONC could look to the FHIR at Scale Taskforce (FAST) work on Scalable Consent Management, for instance, and amplify the solutions industry is working to develop to address a standardized, but modular, approach to consent. It is important to ensure the approach to consent promoted does not significantly add to provider and payer burden.

The Da Vinci membership also supports using the **HL7 FHIR Bulk Data Access IG (Bulk)** for the Provider and Payer Access APIs, as well as a number of other use cases. For these APIs, use of Bulk supports when a provider has a panel of patients associated with a single payer or when multiple patients are identified as moving from one payer to another. In this situation, the payer can share those data asynchronously in one transaction. Use of the Bulk standard also supports sharing data for a single patient asynchronously, which may also be valuable to support data sharing.

There is value in leveraging the Bulk IG across use cases as more and more data need to be moved via FHIR. Helping the community address constraints related to attribution and consent to best support Bulk transactions, and encouraging continued work by the HL7 community to address resource issues experienced when leveraging the Bulk IG, as well as other challenges that have prohibited FHIR Bulk Data Export from being most performant are important opportunities for CMS and ASTP/ONC to advance these use cases.

Provider Directory API: CMS-9115-F also required a Provider Directory API that can leverage the HL7 FHIR Da Vinci PDex Plan Net IG to expose provider directory information via a public-facing endpoint. As discussed further in PR-7, a validated payer/provider directory is critical to the success of interoperability efforts, and this Provider Directory API is a good start to making important information available via FHIR. To be most valuable, it is important that the directory information exposed is current and accurate (see additional discussion on the critical need for a national directory in PA-4).

Prior Authorization API: CMS-0057-F requires use of the Prior Authorization (PA) API and the National Standards Group (NSG) announced enforcement discretion to allow the use of FHIR for PA consistent with this final rule. This API relies on the three HL7 FHIR Da Vinci Burden Reduction IGs:

- Coverage Requirements Discovery (CRD),
- Documentation Templates and Rules (DTR), and
- Prior Authorization Support (PAS).

The Da Vinci Project strongly supports the use of these APIs and the Da Vinci Burden Reduction IGs to facilitate it. As we have noted in previous public comment, PA is increasingly documented as a burden for providers and payers, and the impact on patients is noteworthy. According to research conducted by RTI on behalf of the AHIP[1], electronic PA can reduce the time between when a PA is submitted and a decision is made three-fold, or by an average of about 69%. This means timelier care for patients, and less burden for all parties engaged in the PA process.

To realize the maximum benefit of electronic PA, the PA process must be embedded in a provider's workflow and fully automated. This will enable beneficiaries to engage in meaningful discussions with their Medicare providers at the point of care regarding their options, thereby facilitating more timely care and adherence to care plans. This approach aims to enhance the quality of care, improve health outcomes, and reduce costs.

These Burden Reduction IGs leverage **CDS Hooks** and **Clinical Quality Language (CQL)**. We find both are very valuable and necessary to support FHIR implementation across use cases. We appreciate there have been concerns raised in the past about both, but we have actively worked with the community to address them. We do acknowledge that one of the challenging aspects of implementing the Da Vinci IGs involves payers doing the work to express the data needed for PA decision-making in CQL so the documentation collection process and the process to evaluate PA requests against medical policies can be automated. In the end, this is less a technology issue and more a business transformation issue. But, it is one process that innovative technology may now be able to support. There is an opportunity to leverage emerging AI tools to support this digitization process, and CMS and ASTP/ONC support for using such tools could significantly aid implementation efforts.

The Da Vinci Project requested and received approval for an exception to HIPAA requirements for Da Vinci payers and their trading partners to test the use of the HL7 FHIR Da Vinci Burden Reduction IGs to improve the PA process. The HL7 Da Vinci Exception Report[2] illustrates the value of using these IGs. Though limited by the number of participants, this test showed a drastic decrease in waste. Using FHIR standards helped avoid up to 84% of unnecessary PA requests, clarifying whether a PA was required or

[1] https://www.ahip.org/news/press-releases/new-analysis-shows-benefits-of-electronic-prior-authorization-for-patients-and-providers

[2] https://confluence.hl7.org/download/attachments/113675673/HL7%20Da%20Vinci%20Exception%20Report_Final%20%28June%2025%2C%202024%29.pdf?version=1&modificationDate=1731382945836&api=v2

not and providing the clinician an immediate response directly in the workflow. This removed overhead costs and care delays. Submission response times were reduced to minutes for many items and services. And, ultimately, participants saw a significant Return on Investment by leveraging FHIR, which allowed them to reserve workforce resources for only the most complex PA cases.

The Da Vinci Project has launched a Prior Authorization Implementers Group including both Da Vinci members and non-members to work together to support increased implementation of these IGs; share what is working and what in not working as payers, providers, and vendors work through implementation; and, to collect metrics to better understand the value of using these IGs in a quantifiable way. Maintaining the existing HIPAA-related enforcement discretion supports continuing to be able to work meaningfully toward electronic PA on FHIR. We look forward to continuing to support the community as they work through this implementation in support of CMS-0057-F and evaluate how we can support the use of innovative tools, such as AI, to better facilitate this implementation. It is critical that there is an emphasis on both provider and payer engagement as PA on FHIR will only be successful at scale with both providers and payers actively investing in and adopting this new approach.

SMART on FHIR: The Da Vinci Project strongly supports the use of SMART on FHIR to advance the use of the FHIR standards and implement these IGs across use cases. Da Vinci members are innovating with SMART on FHIR to find ways to work both within EHRs and outside but in support of EHRs to support FHIR implementations of everything from Prior Authorization to FHIR-based questionnaire and assessment data collection and submission. Some of the primary constraints noted for effective implementation of SMART on FHIR app solutions are around integration with a local system and support at the point of care.

Overall, the constraints and challenges experienced to date with implementing these APIs are far outweighed by the opportunity to automate interoperable data exchange at scale, and without these APIs and FHIR-based tools, such automation would not be possible. That said, as we have previously commented, IGs and underlying standards are evolving more quickly than the established regulatory cycle. It is equally important to ultimately require specific IGs to ensure industry has clear direction and a common set of interoperable standards to definitively work from as it is to find the regulatory flexibility in the new world of rapid standards development to allow a new version of an IG to be used as soon as it is available. We recommend that CMS and ASTP/ONC work closely together to establish a clear, predictable path to interoperability. This path should include required IGs, the ability to advance IG versions outside the regulatory cycle, provide adequate time for industry adoption of new IG versions, and limit the IG options available for support at one moment in time as not to disrupt interoperability. The Da Vinci membership looks forward to working with CMS and ASTP/ONC to find the best approach to IG versioning requirements and promoting active industry participation in standards development.

PR-7 – Strategies for High-Quality, Timely, & Comprehensive Data: The Da Vinci Project feels strongly that the primary way to support providers in making high-quality, timely, and comprehensive healthcare data available to facilitate interoperability is to make a validated payer/provider directory available. Across use cases, without a high-quality, FHIR-enabled national directory, implementation will be challenging. All stakeholders engaging in data exchange need to know with confidence how to identify their data exchange partners.

We believe it is critically important to develop a central hub of standards-based healthcare directory information that can then feed downstream systems and directories using FHIR-based APIs. This must be

a truly national directory, including both public and private commercial directory information to support meaningful data exchange across the healthcare ecosystem.

It is important this directory be validated. It is equally important that all FHIR implementations be validated to ensure high-quality, timely, and comprehensive data are being exchanged via defined APIs. Ideally, such validation would be at the implementation level – attesting information is available, accurate, and timely is not good enough. CMS and ASTP/ONC could support implementation-level validation across FHIR use cases by ensuring developers and implementors have access to testing tools during the process of development and as API solutions are being rolled into production. A sandbox environment for testing regulated APIs would also be highly valuable. No matter how standardized an implementation is, there will always be the need for custom code to ensure the API can work within a local system, workflow, and in accordance with necessary policies. A national sandbox that could support real world testing would greatly aid development and testing. Testing often and testing with standardized tools can ensure truly interoperable solutions are being built and deployed. This is the first step in making sure high quality and complete data are being exchanged. Furthermore, we encourage incorporating testing with third-party apps and patient users of those apps to ensure patients receive actionable data from the APIs they can use to be partners in their care.

Metrics are also important. CMS and ASTP/ONC need to be able to measure if implementations are meeting necessary standards of quality, timeliness, and completeness. And, to support collection and reporting of metrics across use cases, CMS could leverage incentives and disincentives across programs to initiate the process of appropriate and complete implementation. Compliance needs to be actively enforced, including API implementation deadlines, and this information needs to be transparently communicated to patients, providers, and payers so that all stakeholders know reliably what information will be available when and in what formats across use cases.

PR-8 – How can CMS or Partners Simplify Provider Clinical Quality Data Responsibilities: The Da Vinci Project believes that leveraging digital quality measures on FHIR can help reduce burden on providers who must collect and submit these data for various purposes to CMS. And, using the FHIR standard can support CMS receiving consistently calculated, standardized quality metrics efficiently and timely. By moving to quality measures on FHIR, CMS and ASTP/ONC could help providers (and payers) leverage the work they are already doing to map data to FHIR and build FHIR solutions across use cases and apply that to quality measurement. Standardized data mappings is one of the critical first steps in improving data quality.

Supporting quality measures on FHIR includes focusing on advancing standards already required in existing regulations such as Bulk FHIR. When submitting quality measure data, it is necessary in certain situations to be able to send data for an entire patient population asynchronously. Only with Bulk is that possible. This will require Bulk to be fully performant.

One of the main benefits of digital quality measures on FHIR is the use of Clinical Quality Language (CQL) to define measures in a computable way. This can drive consistency in measure specification interpretation and automate the measure calculation process, which can lead to better data quality and lower burden. Leveraging FHIR resources such as MeasureReport, the specific use case being implemented can determine the best level at which the data should be shared – final measure calculations, consistently derived through an automated process at a population or patient level or aggregated as necessary.

There is work to be done to support a move to digital quality measures on FHIR. It is important that all stakeholders impacted by the calculation and reporting of digital quality measures engage in a discussion about how to best leverage FHIR so that the benefits of automation and standardization are fully realized while maintaining patient privacy and data security. Ultimately, the long-term benefits and efficiencies of digital quality measures on FHIR are significant. And, leveraging FHIR for clinical quality measures could greatly reduce burden on providers and payers. One Da Vinci member is already seeing a 50% reduction in cost by reporting quality measures on FHIR.

If registries were also required to maintain data on FHIR, providers working with these registries could more easily exchange data that could be used in computable measure calculations and/or the registries could support providers in standardized measure calculations.

3. Digital Identity

PR-9 & PR-10: The Da Vinci Project believes it is important that there is a digital identity solution. Implementation of the Patient Access API, for instance, will not be possible at scale without digital identities. Digital identity is an important step toward enabling trust, but it is not alone sufficient – work related to patient matching is still needed. But, having a digital identity provides patients with a right of access to their data and supports secure and trusted data exchange. However, it is important that CMS and ASTP/ONC do not defer a national digital identity solution to one commercial entity. If digital identity is a proprietary solution, it will potentially limit, not accelerate, access. If a single option is preferred, a single Federal option would not create the same barriers a single commercial option would.

II.D. Payers –

PA-2 – How Can CMS Accelerate Payer Adoption of APIs: The Da Vinci Project feels it is important to support payers in their implementation and hold them accountable for not only implementing the required APIs, but promoting their use. Similar to strategies that can support better and more timely data for providers (PR-7), supplying payers with tools to test during development at the implementation level will help payers more efficiently build truly interoperable and valuable solutions. A national sandbox to test implementation could provide payers with testing partners early in the process. This can not only ensure development is accurate and efficient, but it can also establish relationships that can support production data exchange with willing partners.

Payers need to know that if they are building a FHIR solution, that solution will be used and all other parties necessary to complete the use case will equally be ready to engage. This is an opportunity for CMS and ASTP/ONC to coordinate closely to ensure that all actors engaged in a use case are equally required and/or incentivized to accurately and fulsomely implement the required FHIR solutions.

It is also important that CMS and ASTP/ONC hold payers – and all stakeholders required to implement FHIR solutions – accountable for their implementation and promotion of the completed APIs. As noted in PR-5, taking the Patient Access API as an example, a challenge is that many patients do not know apps are available to them and app developers are not always able to access data that should be available to them. There is an opportunity for CMS and ONC/ASTP to support making FHIR API solutions known using incentives and/or disincentives to encourage payers not just build required FHIR API solutions but encourage and facilitate their use.

PA-3 – Payer Acceptance of Digital Identity Credentials: The Da Vinci Project feels that it would be valuable to have a digital identity across the healthcare ecosystem, as noted in PR 9 &10. Implementation, however, would need to be carefully managed to ensure the chosen solution is affordable, accessible, and does not introduce new barriers. A national standard that is Federally managed would be ideal, as an approach that relies on a single commercial entity may be prohibitively expensive and potentially intrusive given the detailed information identity companies generally require. If the digital identity approach was Federally backed, universally used, and truly secure, we believe it would be accepted and utilized.

PA-4 – Value of a Nationwide Provider Directory: As noted in PR-7, it is critically important to have a national provider directory to support truly interoperable data exchange and to ensure the investment payers (and providers) are putting into their FHIR data exchange solutions add value for patients. It is a significant barrier to FHIR API implementation not to have such a directory. It will be challenging to scale the use of FHIR without it.

Having a national single source of truth will help address three major pain points payers, providers, and patients, are experiencing today: 1) it will remove the burden on providers and payers associated with having to update countless directories and/or verify that information in many places often, 2) it will remove the negative impact on care quality that occurs when directories include inaccurate or incomplete information, such as delays in care when a Prior Authorization cannot be efficiently submitted, and 3) it will allow payers and providers to effectively and efficiently transmit secure, standards-based electronic healthcare information across the care continuum where they previously could not due to unknown digital endpoints.

To support payer to provider and payer to payer data exchange, it is important to at a minimum focus on providers, payers, and their organizational affiliations. Not all individual providers will have an endpoint – for some, the endpoint will be an organizational endpoint. It is therefore necessary to understand provider affiliations. To supply enough information for endpoint discovery, and to support finding providers and payers in a national directory, it would also be valuable to include no less than provider demographic information and provider and payer identity information (such as NPI or TIN) to start. Because the National Association of Insurance Commissioners (NIAC) ID is not a truly sufficient payer ID, it would be valuable to also leverage the SMART App Launch “brands” concept that would make it easier for users to know they are identifying the correct payer when working to identify the appropriate endpoints.

With this basic information payers (and providers) would have what they need to implement the current CMS and ASTP/ONC interoperability requirements and to take them to scale.

PA-5 – Payer Support with Provider Clinical Data Responsibilities: The Da Vinci Project feels that payers can best support providers with clinical quality data responsibilities by making sure the data needed are available and in the required formats. Maintaining the needed data in the appropriate FHIR formats, such as claims data on FHIR, will facilitate exchange through the appropriate APIs and ensure providers can leverage these data for FHIR-based quality measure calculations without significant additional effort. As noted in PR-8, we feel there is great value in moving to quality measures on FHIR. One benefit is that the data, if FHIR-enabled, can be leveraged across use cases. Moving to FHIR helps payers and providers move beyond their legacy systems and the associated processes and think about new ways to efficiently and effectively support each other’s data needs.

II.E. Technology Vendors, Data Providers, and Networks –

1. Ecosystem

TD-1 – How Can CMS Stimulate Developer Interest in Building Digital Health Products: The Da Vinci Project believe that a primary way CMS can stimulate developer interest is by continuing to support and invest in FHIR solutions for Medicare and Medicaid beneficiaries, such as Blue Button, and promote use of consumer-facing apps that leverage the data. CMS and ASTP/ONC could encourage developers to build patient-focused apps and other beneficial digital health products through challenges and competitions that promote the products that best serve patient and program needs. And, CMS could work to integrate use of CMS APIs like Data at the Point of Care (DPC) and the Beneficiary Claims Data API (BCDA) across programs. CMS could also consider opening APIs like DPC and/or BCDA to payers. The more FHIR-enabled data are appropriately available across the healthcare ecosystem, the more developers will be motivated to support use of those data.

As a leading payer in the healthcare ecosystem, CMS can signal to industry the value of investing in these technologies and promote their use. As noted with the Patient Access API (see PR-5), a primary barrier to adoption has been awareness. CMS can not only support internal use of the APIs within their own programs, such as CMS Innovation Center Models, but also promote the apps and digital health products that make the data available via these APIs valuable to patients. CMS and ASTP/ONC could also ensure that developers do not experience barriers to data access and help streamline this access. CMS might consider incentivizing payers and providers for promoting the use of such technologies. For instance, CMS could add Quality Payment Program (QPP) Improvement Activities specifically related to interoperability and demonstrated use of apps and other digital products both by patients and providers.

Finally, ASTP/ONC could continue to leverage Leading Edge Acceleration Projects (LEAP) funding and similar grants to support development. And, CMS and ASTP/ONC could host challenges and competitions to develop value-add apps and digital products leveraging CMS data to encourage development and promote use of high-quality apps and products.

Continued federal engagement and investment in promoting the use of FHIR-based APIs will be instrumental in stimulating developer interest and motivating providers, payers, and vendors to continue to advance these efforts.

TD-2 – Data to Stimulate Developer Interest:

a. The Da Vinci Project believes that data that support the calculation of CMS quality measures would be one type of **data that would be valuable to make available via CMS APIs**. Ensuring providers and payers can send and receive the data necessary to support quality measure reporting via FHIR-based APIs would support FHIR-based quality measure advancement and significantly reduce burden.

In general, it would be valuable for CMS to align the data available via APIs to data required by CMS and other Federal programs. Ensuring that program requirements could be met leveraging FHIR APIs and/or encouraging industry to develop products that can help providers and payers meet requirements using the appropriate data would support the health digital product ecosystem grow because there would be clear use cases that need attention, and it would also result in reduced burden as providers and payers would have better tools to support compliance.

To further advance use of FHIR-based APIs and related technologies, CMS should work to make interoperability part of all programs, starting at program development with interoperability goals and a strategy to use FHIR standards and appropriate APIs and digital products clearly articulated prior to

program start. Da Vinci suggests initiating these requirements with Innovation Center Models and quality reporting programs – two areas most likely to see immediate benefits from leveraging new technologies that support secure receipt and exchange of standardized data.

b. CMS should consider **adding data to Blue Button 2.0** that could help consumer-facing apps truly aid patients, specifically regarding improving their health, following their care plan, care gaps that need to be addressed, and effectively navigating the healthcare system. This could include quality metrics that help them see key health markers (such as A1c level and other chronic disease markers, completion of mammograms or colonoscopies, mental health screening markers, etc.). This could include Medicare Advantage star rating information and other Medicare Advantage metrics to help beneficiaries choose plans. This could include provider directory information leveraging HL7 FHIR Da Vinci PDex PlanNet as the Patient Access API does for the health plans CMS regulates, as well as the HL7 FHIR Da Vinci Member Attribution (ATR) List IG. This could also include price transparency information both related to hospitals consistent with existing regulations and the latest Executive Order 14221 and related to patient cost transparency as defined by the No Surprises Act (see TD-19 for more information on the HL7 Da Vinci Patient Cost Transparency IG and how this could be leveraged to support adding patient cost transparency data to a FHIR-based API for use by a third-party app of a patient's choice).

c. The Da Vinci Project feels that the primary barriers for developers are navigating privacy rules, consent, and data use agreements/data access, specifically for third-party apps and digital products. For the APIs to be most useful, particularly beneficiary focused APIs, the data have to be curated for them in an actionable way. This requires the use of products that need to be able to get the right data for the right patients timely. To spur innovation, it is critical that developers have sufficient access to all data needed and that they are not subject to unnecessary access barriers or costs. Data quality is another barrier. This is particularly true for directory data. For patients to get the most use the data have to be current and accurate. And, key information such as whether a provider is accepting new patients must be current.

d. **Additional APIs that CMS and ASTP/ONC should consider** including in program policy are APIs that support cost transparency, quality reporting, and scheduling. The Da Vinci project is working on the IG to support patient cost transparency consistent with the No Surprises Act (see TD-19). More work is needed to support developing FHIR standards that could support hospital price transparency.

Such significant effort is needed to accurately report quality measures required by so many CMS programs, more support to calculate and submit these measures on FHIR would greatly reduce burden and could be a first step in improving data quality through standardized and computable measure specifications and FHIR-mapped data.

Scheduling APIs could benefit both patients and providers and Argonaut has led work in the HL7 community on FHIR resources that could be leveraged for this. Such APIs could help patients get more timely care, improving outcomes and lowering cost.

3. Technical Standards and Certification

TD-4 – How Can CMS Encourage Open Standards and Publicly Available APIs: The Da Vinci Project believes that CMS can encourage the use of open standards and publicly available APIs by leveraging the ONC Health IT Certification Program to continue to push the support and use of open standards like FHIR and integration of APIs. If CMS programs continue to require the use of this certified technology, they can require the use of FHIR and APIs to meet program goals with the confidence the providers engaged in the programs will have the technology to support the requirements. CMS can also consider leveraging

program-level incentives and/or disincentives to support use of these standards. If use of open standards and innovative health IT is considered at the program development stage (as noted in TD-2), program participants will have ample opportunity to prepare to successfully meet these requirements and advance the use of these standards.

It is important, however, that CMS and ASTP/ONC follow through on validating that such requirements are being met and measure the use and impact of the standards and technology so that industry is held accountable. Again, promotion of these standards and their value, as well as education to ensure patients, providers, payers, and industry understand CMS and ASTP/ONC goals and the benefits of using open standards over proprietary standards will also support their use.

TD-5 – National Provider Directory of FHIR Endpoints: The Da Vinci project feels strongly that a National Provider Directory is critical to the ability to implement FHIR and achieve interoperability at scale. As noted in PA-4, to support payer to provider and payer to payer data exchange, it is important to at a minimum focus on providers, payers, and their organizational affiliations. Generally, individual providers will not have an endpoint – the endpoint will be an organizational endpoint. It is therefore necessary to understand provider affiliations. To provide enough information for endpoint discovery, and to support finding providers and payers in a national directory, it would also be valuable to include no less than provider demographic information and provider and payer identity information (such as NPI or TIN) to start. Because the National Association of Insurance Commissioners (NIAC) ID is not a truly sufficient payer ID, it would be valuable to also leverage the SMART App Launch "brands" concept that would make it easier for users to know they are identifying the correct payer when working to identify the appropriate endpoints.

We believe that a central, federally managed national directory that serves as a central hub that could then feed all downstream directories and systems using FHIR-based APIs is the ideal approach. This is a critical national resource to ensure that the healthcare ecosystem can implement standards-based data exchange at scale to realize all the efficiencies and burden reduction opportunities a FHIR-based system can bring. We appreciate this is an expense, but the money saved by not only consolidating the proliferation of directories across industry, but within CMS alone would be a significant cost savings. A centralized directory would allow payers and providers to come to one place and make all needed updates. Through automation and APIs, the data from this central source could provide the data as needed to downstream systems. The efficiencies gained by using one system would support the cost of building it. CMS and ASTP/ONC could also consider factoring the cost into CMS programs and adjusting incentives and disincentives accordingly. Ultimately, this is a necessary national utility that can support a more efficient and effective healthcare system.

TD-7: The USCDI Benefits and Limitations: The Da Vinci Project believes that the USCDI provides a valuable and necessary foundation for advancing the use of the FHIR standard across the healthcare ecosystem. By setting the floor on which data concepts at a minimum must be supported for FHIR exchange it gives industry a shared starting point. This is valuable.

There are two major limitations of the USCDI. First, it is more of a policy statement than a data model that it is reliant on the US Core IG for implementation details. The necessary implementation details are not fully realized in the US Core IG for all data concepts and so there is room for interpretation, and thus inconsistent implementation, on certain concepts. This can negatively impact interoperability. For instance, disability status is a USCDI data concept, but it is not specifically defined as one assessment or a core set of data elements. As a certified health IT vendor, I may not be prepared to share and receive

the assessment CMS ultimately defines in a program or Innovation Model to collect these data because it is unclear what the requirement to support "disability status" will mean at the operational level based on the USCDI. It will be important to ensure that the HL7 community can continue to work through the operational details so that implementers have the necessary information to utilize the USCDI to support truly interoperable FHIR solutions.

The second limitation is that depending on the system and the use cases it supports, not all data classes in the USCDI may apply, but if health IT is required to certify to and show that every data class in the USCDI can be accommodated by the technology, health IT vendors may be spending time and money to illustrate a capability they will never use because they will never have the data in their systems to exchange. See TD-9 for additional information on how the Da Vinci Project believes this limitation can be addressed.

Despite the limitations of the USCDI, the ultimate goal is to share and receive all the data needed and warranted for the given use case from the full patient record on FHIR and the USCDI helps us start moving in this direction in a strategic way. Structured FHIR-enabled data elements are critical, however, or we risk having siloed solutions that will not permit valuable, actionable data exchange. A smaller set of structured data is more actionable than more data that cannot be meaningfully integrated and used by a receiving system or app.

TD-9 – Benefits and Drawbacks of Certification of Health IT: As previously indicated in past RFI comment responses, the Da Vinci Project believes that there is value in health IT certification in that certification can ensure health IT products are compliant with the standards and support interoperability. It is also critical to ensure that health IT certification is not overly burdensome.

The Da Vinci Project believes there is a need to "right-size" health IT certification for the health IT being certified. Health IT must certify to every part of a required IG exactly as written. When thinking about the USCDI, operationalized by the US Core IG, certain health IT may not have data to populate every data class, and that health IT instance may not currently be required to exchange the data in question per the CMS API requirements that would ideally align with health IT API certification criterion. But, to certify, the health IT instance would have to demonstrate that it can support all data even if the data are irrelevant for the use cases the health IT is built to support. This adds significant burden for the purposes of certification without adding real benefit for future interoperable data exchange. Providers and the vendors who support them are experiencing this burden today.

There is an opportunity to address this burden. As an alternative, the appropriate health IT systems could certify to the most relevant data classes for their use case for the given API. This could both ensure interoperability and "right-size" certification for each type of health IT system, supporting beneficial outcomes while limiting burden and cost. There is also an opportunity to continue to expand the USCDI + initiative so that health IT systems could certify to the best set of core data to support interoperable data exchange for the use cases it is built to support without having to certify to every data element in the US Core IG.

This approach would also help address concerns related to health IT having to certify to a specific version of an IG. In the HTI-2 Proposed Rule (RIN 0955-AA06), ASTP/ONC called out several Da Vinci IGs to support the newly proposed certification criterion related to patient, provider, and payer APIs, which the Da Vinci community appreciates. We also appreciate the work done in that proposed rule to align with the CMS interoperability requirements called out in the proposed criterion – specifically the Patient Access API, the Provider Access API, the Payer-to-Payer API, the Provider Directory API, and the Prior Authorization API certification criterion. However, we raised significant concerns with requiring specific

versions of the IGs. Currently, the relevant CMS regulations – the CMS Interoperability and Patient Access Final Rule and the CMS Interoperability and Prior Authorization Final Rule – recommend certain IGs be used. This strong recommendation has been understood by implementers as a clear direction and thus we are seeing the growth of interoperable implementations. Not locking in a specific version of these IGs while they are still maturing provides an opportunity for the IGs to evolve and improve and for implementers to continue to innovate.

At this stage of evolution of these IGs, strongly recommending the IGs for certification generally, and signaling that as these IGs mature and as additional real-world testing and implementation help these IGs stabilize, specific versions will be required for future certification will serve to help move the industry toward greater standardization and interoperability. Simultaneously, it will provide payers, providers, and vendors the opportunity to more quickly advance the standards and ensure they will truly meet intended needs without taking significant effort to certify to a version that may no longer be relevant or best suited for implementation. This adds significant burden just to get certification without benefiting future interoperable data exchange.

ASTP/ONC did propose to use the Standards Version Advancement Process (SVAP) to help IG versions keep pace with advancements in the industry, and this is welcome and necessary. However, there is concern that the annual nature of the SVAP process and the time between when a new version of an IG might be accepted via SVAP and when that version is officially required can be significant per what we see today with the USCDI SVAP process. This creates challenges regarding how to manage supporting multiple versions of an IG at once and knowing clearly which version a health IT system must certify to versus what might be the version most appropriate or necessary to implement. It is important to note that at this stage of these IGs in their development, new versions can include breaking changes and, in some cases, updated versions are needed because an older version is simply not implementable in all aspects as written. It would not support interoperability and patient access or reduce burden if a health IT system had to certify to a version that is not implementable simply because it was the version named in the regulation.

Also, requiring health IT to certify to a specific IG may stifle innovation. For instance, there are currently IGs that require using CDS hooks, but implementers are currently evaluating if an operation versus a CDS hook could best execute the needed action. By requiring health IT to certify to CDS hooks as written and not provide options to explore innovative approaches that do not impact the interoperability of the ultimate data exchange, the pace at which the community is able to advance and test how best to improve and iterate on an IG may be slowed. Given the evolving nature of the IGs today, having to certify to a single version at a defined regulatory cadence could negatively impact IG development and advancement and add burden.

We do feel strongly that if ASTP/ONC required the use of FHIR R4 for an API certification criterion this would ensure FHIR-based solutions are used that start by leveraging the recommended IGs but would leave the necessary room for the community to more quickly advance the standards and move between versions as needed to ensure adoption and implementation. ASTP/ONC could also consider requiring these APIs as functional criterion with a recommended set of IGs to start. In the future, we would support requiring specific versions of IGs that are more stable. And, if possible, we would recommend the “major” version of the IG is required versus the “minor” version – meaning ASTP/ONC require “version 3.1” versus “version 3.1.1”. This would provide added flexibility for implementors to accommodate technical corrections without having to wait for the SVAP process or future regulation.

The Da Vinci Project and its members are committed to continue to work with the healthcare community to advance FHIR standards to benefit all patients, providers, payers, and vendors, and we look forward to the continued opportunity to work with ASTP/ONC in an effort to formulate a next-generation health IT

certification program that best serves the healthcare community and advances true innovation and interoperability.

TD – 10 & 11 – Advancing Data Exchange Without Special Effort for a Patient's Full Health Record and Supporting EHI Export: The ultimate goal is to have the full patient's record available so that the right data can be available at the right time in the right place. If the full patient's record is available per applicable regulation and law, a payer could access the portion they need, a provider the portion they need, and digital health products the portions necessary to meet user needs and product goals. For this to be most valuable, all data in the patient's record would be FHIR enabled and standardized. We are not yet there. Only a portion of the data, largely that defined within the USCDI, is standardized. But, that is a valuable start.

The Da Vinci Project believes that ASTP/ONC should support certified health IT's ability to share as much data about the patient as is needed as we work to get to a fully FHIR-enabled patient record facilitated by a comprehensive USCDI. To do this, it would be best to leverage Bulk to transport data in an existing format that is more actionable to the receiving system, such as C-CDA. The challenge with EHI Export is that it is unstructured data that is hard to use. Very few systems can truly work with a full patient record, especially if unstructured. It is so much data and requires manual review, which increases burden.

Leveraging Bulk and existing formats such as HL7 C-CDA could be an interim solution. Ultimately, we want to ensure that only the minimal data needed to best serve the patient and meet the use case at hand is accessed. To do this, we need to get to a fully structured patient record. We are incrementally working our way toward that end by starting with the USCDI – the structured subset of the patient record.

As with many use cases, the main constraints to date for large data exchange are issues with compute resources that have limited Bulk data transfers, as well as issues with consent and attribution when looking to share large amounts of data across a panel of patients. ASTP/ONC could continue to support industry in working through concerns and considerations around consent and CMS and ASTP/ONC could continue to support the HL7 community is addressing attribution and Bulk implementations to help facilitate this data exchange (see TD-15 for further considerations).

4. Data Exchange

TD-12 – CMS Endorsement of Non-CMS Data Sources and Networks: The Da Vinci Project believes that it is critical that all data that help address whole person care should be available as needed across the healthcare ecosystem. This includes all clinical and administrative data as well as resources such as Community Resource Network data that can help connect healthcare and social needs bringing together public and private resources to fully support patients. The healthcare ecosystem is an interconnected web of public and private supports and services. As much as CMS can help digital health IT leverage all the available data, the better providers and payers will be able to best serve the whole patient and truly ensure better health outcomes at lower cost. To support value-based care, CMS and ASTP/ONC may want to consider starting with post-acute and ancillary services data which can provide valuable insights that support managing total cost and improving outcomes.

TD-13 – Opportunities and Advancements as a Result of API EHI Access: As noted in TD 10 & 11, the ultimate goal is to ensure that whatever data are needed on a patient is accessible when and where the data are needed to ensure that patients are getting the best and most informed care possible. This supports sharing as much data as are available via FHIR-based APIs even if the data are not yet FHIR enabled. To do this, it would be best to leverage Bulk to transport data in an existing format that is more actionable to the receiving system, such as C-CDA. The challenge with EHI Export is that it is unstructured data that is hard to use.

The Da Vinci Project believes it is important to facilitate sharing the data needed, even if we are still working our way toward a full patient record on FHIR. With more experience sending and receiving all the available data, we will learn how to refine what is shared, how to most strategically move toward greater standardization, and how to technologically overcome compute resource obstacles, as well as how to overcome attribution and consent obstacles from a technology and policy perspective. CMS and ASTP/ONC could support more effective data sharing and use of Bulk data sharing through pilots or through incorporating such data sharing in Innovation Center Models. This will particularly help industry work together with CMS and ASTP/ONC to understand the necessary infrastructure investment and data governance considerations related to managing exchange of the full patient record.

TD-14 – Networks' Use of FHIR APIs: Da Vinci Project members will share more specific information about their use of FHIR APIs, but as a community, we believe it is important when thinking about networks' use of FHIR APIs to reiterate the need for a national directory, a non-proprietary central source of truth that can provide all stakeholders within the healthcare ecosystem the endpoint information needed for interoperable FHIR API data exchange at scale (see PA-4 and TD-5). If providers and payers and the health IT products that support them and their patients can identify the correct FHIR endpoints, then standardized data exchange can occur. A national directory is a foundational need to support cross network use of FHIR APIs. Another foundational need is digital identity and ID verification, as discussed in PR-9 & 10 and PA-3, to ensure that the right data for the right patients are being shared across networks.

TD-15 – Advantages and Disadvantages of Bulk FHIR APIs: The Da Vinci Project feels that there is great value in continuing to advance the use the Bulk FHIR APIs. As discussed in PR-5, Bulk FHIR is necessary to support the Provider and Payer Access APIs at scale. As an increasing number of use cases are identified where it would be most efficient to share data on a panel of patients, Bulk FHIR is critically important. It can support everything from population health data exchange, to research, to quality measures on FHIR. It can support ensuring data are captured once and are then available for all authorized use, consistent with a critical objective of the 21st Century Cures Act. Asynchronous FHIR-based data exchange is necessary to ensure a variety of use cases can be supported.

The primary disadvantages are that members of the community are experiencing issues with compute resources when trying to leverage Bulk as many EHR instances are still hosted on prem versus on the cloud. It would be valuable for CMS and ASTP/ONC to consider if it is always the EHR that needs to support Bulk or to rethink how to ensure that non-EHR provider systems can support Bulk at a performant level without compromising their users in cases where the EHR is not suited to support such a data exchange. Another obstacle has been how to best manage consent and attribution for Bulk FHIR data exchange. The Da Vinci community has been evaluating how the Da Vinci Member Attribution (ATR) List IG can help address attribution concerns. CMS and ASTP/ONC could consider requiring patient search, including advanced searching like the _typeFilter parameter, and group IDs to support better use

of Bulk FHIR. We encourage CMS and ASTP/ONC to continue to work with the HL7 community to work through these issues across Bulk use cases. Bulk must be fully performant to truly support FHIR API adoption at scale across use cases.

5. Compliance

TD-19: Implementing Price Transparency: The Da Vinci Project has been directly engaged in working to address price transparency needs through our work on the HL7 FHIR Da Vinci Patient Cost Transparency (PCT) IG in support of the No Surprises Act (see Division BB, Title I, Sections 111 and 112)[3], which was enacted as part of the Consolidated Appropriations Act, 2021. The No Surprises Act specifically requires that a provider or providers share a Good Faith Estimate (GFE) with a patient when the patient is self-pay/uninsured or with a payer for insured patients when the patient schedules a service or requests an estimate. When the payer receives a GFE, the payer must make an Advanced Explanation of Benefits (AEOB) available to a patient in advance of the service.

Understanding the need for multiple providers to be able to work together to produce a single GFE in support of the law, this IG provides guidance on how to create this GFE using FHIR. This IG introduces a GFE coordination workflow that leverages a Coordination Platform. The Coordination Platform could be any number of systems - a practice management system, an EHR, a payer-supported system, a third-party system such as a cost estimator, billing services, clearinghouse, etc. When a patient contacts a provider to schedule a service or request an estimate for a service, the provider or facility who schedules the service or receives the request is the convening provider or convening facility per current CMS regulation for uninsured/self-pay patients. The IG provides the tools to kick off a workflow leveraging FHIR when the convening provider or facility can contact all the necessary co-providers, as needed, support the creation of one comprehensive GFE, and a payer-supplied AEOB for the patient. The IG also supports a mechanism for this information to be shared back with the patient via FHIR, including an option to include these data in the Patient Access API so a third-party app of the patient's choice could make these data accessible to them in a user-friendly and actionable way.

The years of work on this use case have illustrated how challenging this work is. It is challenging for providers to coordinate to produce a single GFE for a full suite of services. It is challenging for payers to estimate the cost of items or services when the patient responsibility could change as new services are rendered and claims are finalized that impact their deductible or out-of-pocket maximum expenses. Leveraging FHIR helps mitigate some of the challenges by supporting automated processes that can support more efficient coordination and faster processing that can help payers and providers meet the legislatively defined timing requirements. FHIR can also help patients get this information more quickly via an app so they are not waiting for a letter in the mail or required to log into one or more portals to access the information.

The Da Vinci Project is thankful for CMS and ASTP/ONC's support for this work. We do believe that continuing to focus on a FHIR-based solution will best help industry meet the legal mandate and reduce burden as many of the FHIR Resources profiled in the PCT IG are the same that are used in the CARIN for Blue Button® IG and the Da Vinci PDex IG consistent with requirements for the Patient Access API (CMS-9115-F) and the Interoperability and Prior Authorization Final Rule (CMS-0057-F.)

We appreciate that there are parallel efforts looking at how legacy electronic data interchange (EDI) solutions could be leveraged to support patient cost transparency. APIs are generally considered an improvement from legacy EDI because they offer greater flexibility, real-time data exchange capabilities,

[3] https://www.ftc.gov/legal-library/browse/statutes/no-surprises-act-2021-consolidated-appropriations-act

and easier integration with modern applications. EDI tends to be more rigid, focused on batch processing, and requires specialized knowledge to implement and maintain. APIs are therefore a more adaptable choice for today's dynamic healthcare ecosystem.

A FHIR-based API approach to sending GFEs from providers to payers would not replace the HIPAA-mandated X12 transactions for claims processing. Claim processing is highly complex, involving multiple parties. To our knowledge, however, the adoption of X12 predetermination claims and cost estimates utilizing portions of the existing claims processes remain significantly lower and less rigorously evaluated. Using a FHIR API to create a GFE could mitigate confusion surrounding the processing of advanced estimates of services by establishing a clear distinction between these estimates and actual, payable claims for payers.

FHIR APIs also enables sharing GFEs with patients, which provides the ability for digital health IT solutions to use the same GFE process for self-pay, uninsured, and insured patients and enables consistent transparency and accessibility of AEOBs and GFEs via API to a patient's app of choice.

The healthcare community is still awaiting tri-agency rulemaking and guidance regarding operationalization of the No Surprises Act for insured patients. While early adopters continue to make progress, clear direction from the Federal government on the requirements will motivate the healthcare community to further engage and invest in implementation. CMS and ASTP/ONC could further support engagement and implementation by incorporating use of price transparency APIs in relevant programs or certification.

The Da Vinci community is grateful to the tri-agencies for supporting a collaborative dialogue across all stakeholders working on digital solutions for price transparency, and we are eager to continue the conversation to identify the best solutions for providers, payers, and most importantly patients. Additionally, we look forward to the opportunity to support conversations on how the work that has been done to develop a FHIR-based solution for patient cost transparency could be leveraged to support cost transparency for additional use cases, including hospital price transparency.

F. Value-Based Care Organizations

1. Digital Health Adoption

VB-3 – Essential Health IT Capabilities for Value-Based Care Arrangements: As an HL7 Accelerator, the Da Vinci Project is responsible for supporting industry, specifically providers and payers, but also critical vendor partners, to develop and implement value-based care standards. We believe that to achieve CMS' value-based care goals, it is necessary to leverage FHIR-based APIs. FHIR-based APIs can support Accountable Care Organizations (ACOs) and any value-based care arrangement with foundational work from supporting patient care to operational workflows to quality measurement. Data are the lifeblood of value-based care – it is necessary to efficiently and effectively access data to monitor and evaluate care and services to assess that the value-based care arrangement is meeting the needs of the patient while also improving overall health outcomes at lower cost. FHIR-based APIs can both facilitate efficient care and support automated analysis of patient outcomes, utilization, and expenditures.

Digital health IT can support value-based care by ensuring the right data are in the right place at the right time through use case specific APIs. ACOs could benefit from leveraging both the Provider Access API and Payer-to-Payer API to ensure the patient's record is available for care delivery support and effective care coordination. The Prior Authorization API can support more timely care and care plan adherence. A

FHIR-based national directory leveraging a national digital identity could ensure the right records are shared with the right entities efficiently to ensure maximum value and benefit from FHIR API use. This is foundational to establish trusted exchange. ACOs already have access to Medicare claims data via the Beneficiary Claims Data API (BCDA). Access to claims data via APIs like BCDA and Data at the Point of Care (DPC) could further support access to claims data, which can support utilization and expenditure monitoring and evaluation on FHIR.

Patients would benefit from access to Blue Button/Patient Access API-fueled apps so they could monitor their own health data and be true partners in their value-based care journey. As CMS works to further incorporate patient reported outcome measures (PROMs) into value-based care programs, digital health IT could support digital monitoring and data sharing leveraging everything from wearable technology data to SMART on FHIR enabled apps and assessments that can capture – in the moment – information to contribute to robust PROMs data collection and submission that is both secure and timely.

There are many opportunities for value-based care arrangements to leverage FHIR and FHIR-supported digital health IT, and doing so is necessary if value-based care is going to truly scale to all patients. These programs also offer an opportunity for CMS to provide direct support and/or incentives to help ACOs build the needed digital infrastructure and adopt digital health technology more broadly and effectively. The Da Vinci community looks forward to the opportunity to work with CMS and ASTP/ONC to continue to advance the use of FHIR in value-based care.